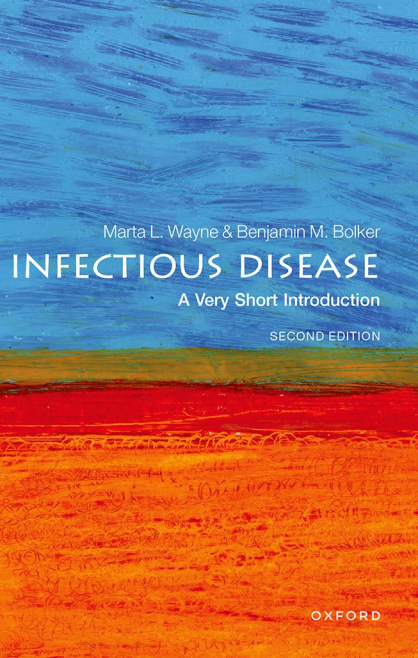




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INFECTIOUS DISEASE

A Very Short Introduction

SECOND EDITION

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Chapter 1

Infection is inevitable

We talk about ‘infectious lyrics’ and ‘viral videos’. Metaphors of infectious disease saturate Western popular culture in the 21st century. The COVID-19 pandemic of the early 2020s has focused the world’s attention on infectious disease, but it is only the latest infectious disease of the 21st century (after SARS, H1N1 influenza, Ebola, and Zika) and it will not be the last.

As recently as the 1970s, doctors were boldly proclaiming the beginning of the end for infectious disease. They thought their arsenal of vaccines for preventing viral diseases and broad-spectrum antibiotics for treating bacterial infections could handle any threat. But disease was never dead, or even in remission. Even as the doctors announced victory, drug resistant strains of *Staphylococcus aureus* (one of the ‘flesh-eating bacteria’ of British tabloids) were spreading in hospitals. (Japan had experienced outbreaks of drug resistant bacteria in the 1950s, but at the time these epidemics were little noticed in the West.) Things got worse as HIV, which has stubbornly resisted the development of vaccines to the present day, emerged in the 1980s. In recognition of the renewed threat of infectious disease, the US Institute of Medicine coined the phrase ‘emerging and re-emerging diseases’ in the early 1990s.

An infectious disease is one that you can catch from another person or organism as a result of the transmission of a biological

agent. In contrast, you fall ill with non-infectious diseases—heart disease, diabetes, Alzheimer’s—because of a combination of your environment and genes inherited from your parents. The agents that cause infectious disease are called *pathogens* or (more broadly) *parasites*. While biologists once used ‘parasite’ only to describe relatively large disease-causing agents such as tapeworms or ticks, they now include microorganisms (viruses, bacteria, fungi, and protists) in this category as well, making the two terms more or less synonymous.

Infectious disease frightens us precisely *because* it is infectious. Its agents are invisible to the naked eye and thus largely unavoidable, except by entirely eschewing human contact. Edgar Allan Poe illustrated the fear of infection, as well as the futility of cutting off human contact to evade infectious disease, in his 1842 story *The Masque of the Red Death*. In Poe’s story, a group of wealthy nobles withdraw to an isolated location to escape from a plague called the Red Death. Ultimately, a costumed stranger infiltrates the group at a masquerade ball. Despite their precautions, the entire group succumbs to the disease.

However ineffective it might be, for most of human history the strategy taken by Poe’s nobles—avoiding disease transmission—has been the only way to combat infectious disease. As the COVID-19 pandemic has shown, this strategy remains necessary even in the 21st century. Since the mechanisms of disease transmission were unknown until the mid-19th century, all human societies could do in the face of an epidemic was to cut off contact with infected areas. In 1665, Isaac Newton retreated to the countryside to avoid the Great Plague of London—and incidentally invented calculus and discovered the law of gravity. In the same year, the English village of Eyam voluntarily quarantined itself to prevent the spread of the plague, with half or more of the villagers ultimately dying. The word ‘quarantine’, which now describes the compulsory isolation of potentially infected people to avoid transmission to others, is derived from ‘quaranta giorni’, the 40 days that

ships had to wait outside the city of Venice to be sure they were free of plague.

Quarantines (at least those better than the one in Poe's story) do block transmission, but they are fundamentally reactive—they are only imposed once we become aware of a serious threat of disease. They help only healthy people living in uninfected populations, not individuals who have already been infected or uninfected people unlucky enough to be stuck in the quarantine zone. On the other hand, quarantines can be effective against any disease, provided that we know something about its mode of transmission (since plague is generally spread by rat fleas, preventing communication among humans while allowing rats to move freely is useless).

Quarantines are deployed to protect groups of people, rather than individuals. As medical science improved, public health officials began to shift their focus from the protection of populations to the protection of individuals. Immunization—the process of protecting people by stimulating their immune systems with foreign substances such as mild pathogen strains or toxins—was the first of several major breakthroughs in individual-focused infectious disease control. Immunization for smallpox was widely practised in Africa, China, India, and Turkey by the early 18th century. It achieved public visibility in the West following its importation to England in 1721 by Lady Mary Wortley Montagu, the wife of the British ambassador to Turkey, and more spectacularly via an 'experiment' in the same year promoted by clergyman Cotton Mather of the colonial city of Boston, Massachusetts. In the face of a smallpox outbreak, Mather and his medical colleague Zabdiel Boylston promoted immunization rather recklessly, against the will of the majority of his fellow Bostonians. Despite several deaths, the experiment demonstrated an effective alternative to physical isolation: immunization protected individuals from infection without restricting anyone's freedom of movement.

Mather and Boylston's experiment also illustrated an ethical conflict between controlling disease for the benefit of an individual and controlling disease for the benefit of an entire population. Bostonians who were successfully immunized were safe from disease, but for several days following immunization they could have transmitted the disease to unprotected individuals. Most modern immunizations involve non-infectious substances, so this particular problem is of lesser concern today, but the conflict between individual and public health, and between individual rights and public health, is very much alive.

Most immunizations can only prevent healthy individuals from becoming infected, or reduce the impact of infection, not cure infected people. Individual-level control of disease took another leap forward in the mid-20th century with the advent of antibiotic chemicals. First derived from common household moulds, bacteria, and even fabric dyes, antibiotics could be used to cure individuals who were already infected. The possibility of curing disease also lessened the fear of quarantine, which had previously been seen as a death sentence.

Between antibiotics to cure harmful bacterial infections and a wave of new and effective vaccines to prevent diseases such as polio, measles, and pertussis, an infectious disease-free future must have seemed within reach to the public health officials of the 1970s. However, public health officials were quickly faced with proof that individual-level control fails for many diseases. Vaccines work by priming the human immune system, and thus they are much harder to develop for disease agents such as malaria or HIV that have evolved strategies for evading the immune system. Antibiotics are only effective against bacteria, not other microorganisms such as viruses or fungi (while antiviral and antifungal chemicals do exist, they are much less broadly effective than antibiotics). With the realization in the late 20th century that infectious diseases were not vanquished after all, research began to shift back towards population-level control.

So far we have divided treatments according to whether they primarily help populations (quarantine) or individuals (immunization/vaccination, antibiotics). Looking more closely, however, we can see that both antibiotics and immunization do help protect populations, as well as benefiting the individuals who receive treatment. Using drugs to cure sick people reduces the impact of infectious disease, because people who recover also stop infecting others. Thus, treating sick people can reduce transmission. Using vaccines to protect people from infection means that some potentially infectious contacts (activities by infected people such as sneezing or sexual activity, depending on the disease) are wasted on people who are protected from disease, again reducing transmission. This so-called *herd immunity* reduces the size of an epidemic even beyond the direct effects of vaccination. If we immunize enough of the population, we can reduce transmission sufficiently to stamp out an epidemic. If we can do this at a global scale, then the disease will become extinct (as in the case of smallpox, one extinction that doesn't bother environmentalists).

If the problem were just that some diseases are harder to control than others, we would still be making progress, albeit slowly, in the fight against infectious disease. Modern molecular biology has provided us with a variety of new antiviral drugs, and vaccines are in development even for such difficult cases as malaria and HIV. But both humans and infectious disease agents are living organisms, and all living organisms undergo ecological and evolutionary change, making infectious disease a moving target. Our growing recognition that we (and our plagues!) are tied to the wheel of life, and our realization that individual-level approaches have failed to free us from the wheel, drives the shift in infectious disease research today.

Ecological processes

As much as we try to deny it, humans are subject to the laws of ecology. We control most aspects of our environment. Motor

vehicles have replaced large predators as the leading category of violent death; we have wiped out most of our potential competitors; and we have domesticated the organisms below us in the food chain. But infectious disease still connects us to ecology's global web.

The most important disease–ecology connection is *zoonosis*, the transmission of new diseases from animal *reservoir hosts* to humans: COVID-19 may be the most spectacular example, but many of the new and emerging diseases that we haven't got a handle on come from animals: Ebola, SARS, avian (H5N1) influenza, and hantavirus are a few of the better-known examples. In fact, almost all infectious diseases originate in this way, and most emerging disease threats are zoonotic. Since it is difficult to vaccinate or design drugs against unknown diseases, this parade of new threats is terrifying: we don't know when the next 'super-disease' might emerge.

Zoonoses are as old as humanity. Smallpox is thought to have moved from rodents into humans at least 16,000 years ago; measles probably moved from cattle to humans when humans first started to live in large cities; and HIV jumped from monkeys and chimpanzees to humans in the early 20th century. However, rapid human population growth and changes in land use have increased human–animal contact, whether in the tropical rainforest (HIV and Ebola) or in the temperate suburbs (Lyme disease).

As well as coming into more contact with animals, humans are moving around the planet at an ever-increasing pace. Contact between individuals, and thus transmission, can happen much faster and over much greater distances than when Isaac Newton moved to the countryside to escape the plague in the 17th century or when Poe penned *The Masque of the Red Death* in the 19th century. Diseases that had previously been confined to narrow regions (generally low income countries) can rapidly expand their ranges. This is true not only for human infectious diseases,

but also for diseases affecting other species whose infectious agents are transported by humans in our luggage, in the food with which we sustain ourselves during travel, or on our shoes. Human travel and commerce can spread disease indirectly, by transporting *vectors*: animals (especially insects) that transmit disease from one organism to another. For example, the international trade in used tyres is spreading *Aedes* mosquitoes, the vector of dengue fever. As well as vectors, we sometimes move the reservoir hosts of zoonoses. The first human infections of the Ebola virus outside Africa, in 1989, came from monkeys (crab-eating macaques) that had been imported from the Philippines for animal experimentation: luckily, the particular strain involved (Ebola Reston) turned out to be harmless to humans.

Increasing movement spreads vectors and hosts to new areas; environmental change allows them to thrive in their new homes. With global climate change, animals and especially temperature-sensitive insects can invade new areas in temperate regions. Although the topic is still controversial, many climate scientists and some epidemiologists are convinced that mosquito-borne diseases like dengue and malaria are already spreading to new populations under the influence of regional climate change. An even greater impact comes from more localized environmental changes driven by human patterns of settlement and economic activity. For example, the larvae of dengue-transmitting mosquitoes thrive in water bodies as small as used tyres and household water tanks. More generally, as people move from rural to ever-growing urban environments, they face greater sewage problems (spreading cholera and other water-borne disease) and encounter new and different kinds of disease-bearing insects.

Evolutionary processes

Ecology constantly exposes us to new epidemics, but evolution is even worse: the diseases we already know change even as we attempt to come to grips with them. As living organisms fighting

for survival, infectious diseases don't accidentally escape our attempts to control them. They are groomed by natural selection to escape. Infectious disease is a moving target that moves faster the harder we try to hit it. Disease biologists frequently invoke Lewis Carroll's Red Queen from *Through the Looking Glass*, who said: 'it takes all the running you can do, to keep in the same place'.

For every disease prevention strategy, infectious diseases have an evolutionary countermeasure. Bacteria did not evolve antibiotic resistance in response to human antibiotic use: scientists have found antibiotic resistance genes similar to modern variants in DNA extracted from 30,000-year-old frozen soil. This isn't surprising, because humans did not invent most antibiotics. Rather, we borrowed or co-opted them from fungi, which had evolved them as a strategy for combating bacteria. However, the widespread use of antibiotics in both medicine and agriculture has allowed bacteria that are resistant to one or more types of antibiotic to outcompete their susceptible counterparts. Other organisms, such as the protozoans that cause malaria, have also evolved resistance to the drugs used to treat them. And when HIV patients are given a single drug rather than a multi-drug 'cocktail', the virus evolves drug resistance within their bodies in just a few weeks. Pathogens evolve resistance to vaccines as well as drugs, but in a different way. Rather than resistance genes spreading within the pathogen population, *strain replacement* occurs—previously rare types that are immune to our vaccines, such as the Omega strain of SARS-CoV-2, take over the population.

Although mosquitoes have smaller populations and lower birth rates than bacteria and viruses, and hence evolve much more slowly, they too have found evolutionary countermeasures to our disease control strategies. In high income countries, DDT use was discontinued as Rachel Carson and others spread the alarm about its harmful effects on wildlife, but vector control strategies based on DDT were short-lived even in middle and low income

countries because DDT resistant mosquitoes evolved within a decade of the onset of mass spraying programmes.

Every aspect of infectious disease biology, not just the ability to resist or circumvent control measures, is constantly evolving. Biologists have noted that the virulence of a disease—how harmful it is to its host—is an evolutionary characteristic of the pathogenic organism. Typically mild diseases can suddenly acquire mutations that make them much nastier. A small number of mutations in the West Nile virus (WNV) that arose in the late 1990s made it far more lethal to birds and mice (and probably humans, although it's hard to know for sure since we don't experiment on humans).

Although mutations are random, evolution by natural selection is not: once pathogens mutate, their subsequent success depends on ecological conditions. Biologist Paul Ewald was among the first to point out that changes in pathogens' ecological conditions, such as a shift from direct person-to-person transmission to water-borne transmission, could favour more virulent forms of infectious diseases. The rise of global air travel may drive evolutionary as well as ecological changes in disease: some biologists have pointed out that mixing between spatially separated populations can encourage virulence, although so far there are no verified real-world examples of this phenomenon.

Outlook

Given these challenges, the elimination of infectious disease—the siren song of the 20th century—seems hopelessly naive, and approaches based solely on protecting individuals appear both untenable and unjust, given inequality in access to healthcare. It would seem that we must learn to live with infectious disease, rather than eliminate it. However, we must also strive to reduce the misery caused by infectious disease. Accordingly, this century has seen a shift from attempts to eliminate the agents of infectious disease, to attempts to understand, predict, and manage infectious

disease transmission at the population level. Synergistic approaches, not simply magic bullets, are required for a sustainable approach for everyone to be able to live well with infectious disease: we must make use of tools from molecular biology, economics, and sociology, among others. We will touch on many of these topics, but focus our book on the disciplines of ecology and evolution: ecology, because understanding ecological relationships helps us understand cycles of transmission; evolution, because disease agents evolve, both on their own and in response to our efforts to control them.